

MOHD ANAS

IIIT Vadodara, Gujarat, India

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Education

IIIT Vadodara

August 2023 – May 2027

B.Tech in Information Technology | GPA: 7.46 / 10.0

Gujarat, India

Relevant Coursework: DSA, Object-Oriented Design, Operating System, DBMS, Software Engineering

Experience

Aarogya Kaya

July 2025 – January 2026

Software Engineer Intern

Remote

- **Architected production backend services** using CQRS, PostgreSQL, and Supabase RPCs, securing **15 RPC functions and database tables** with **JWT/RBAC and Row-Level Security**.
- Improved **read performance for a service serving 1,000+ daily users**, cutting average latency from **420ms to 38ms** via in-memory caching and Redis.
- Built **CI/CD pipelines** running automated **unit and E2E test suites** (Playwright), linting, and build validation across pull requests and merges.

Open Source

July 2025 – May 2026

Software Contributor

Remote

- Migrated the **Next.js (Vercel) patch-next utility** from CommonJS to TypeScript ESM, saving **15–30 seconds** per local reinstall during development and testing.
- Contributed **22 merged pull requests and 12 issues** through **maintainer reviews, design discussions, and iterative feedback**, fixing bugs, implementing features, and improving developer tooling.

Projects

A3.store | Python, AWS EKS, Kubernetes, Terraform, gRPC

- Built a **leaderless distributed key-value store** with **gRPC replication, quorum consistency, gossip-based failure detection, and hinted handoff** across a **3-node, RF=2 cluster**.
- Deployed **3 stateful replicas on AWS EKS** with Kubernetes/Terraform, implementing autoscaling, readiness/liveness probes, and rolling updates.
- Designed a **Pytest automation suite** covering storage, replication, and failure-recovery paths, enforcing coverage gates in **GitHub Actions**.

Second Brain | Python, FastAPI, Redis Queue, LLMs

- Architected a **multi-stage AI reasoning backend** that transforms natural-language decisions into **persisted scenario graphs, tradeoffs, branching outcomes, and action plans** through pass-based LLM orchestration.
- Built asynchronous execution with **FastAPI and Redis/RQ background workers**, hardening LLM workflows with **multi-model fallback, structured-output parsing, malformed JSON recovery, and asynchronous HTTP client reuse**.

Technical Skills

Languages: Go, Python, C/C++, Java, JavaScript, SQL

Backend & Systems: REST APIs, gRPC, PostgreSQL, Redis, Distributed Systems, System Design

Cloud & DevOps: AWS, EKS, Kubernetes, Docker, Terraform, GitHub Actions, CI/CD

Testing & Observability: Pytest, Playwright, Unit Testing, Integration Testing, E2E Testing, Prometheus, Grafana

AI & LLM: LLM Orchestration, RAG, Agentic Workflows, Prompt Engineering, OpenRouter

Achievements

- Ranked **365th** among **15,804** participants in CodeChef **START192D**
- **3,026th** among **17,000+** participants in Codeforces Round 1110 (Div. 1 + Div. 2).

Leadership & Activities

IIIT Vadodara

Joint Secretary, DOT Club / Technical Committee Core Member

Gujarat, India

- Organized an intra-college hackathon and mentored **25+ teams** as a **Technical Committee Core Member**.
- Led recruitment, mentorship, and event initiatives for **100+ students** as **Joint Secretary of DOT Club**.